



From lecture session 10: We explored the basic properties of water (like density) and how the stuff dissolved in seawater (mainly salts) have important impact on such properties. Here we will understand what is dissolved in the ocean water, its sources, and proportions.

Lecture Session 11

1. Describe how water is able to dissolve stuff (Hint: Polar molecules);
2. Seawater constituents and sources;
3. Ions in Seawater their relative proportions;
4. Conservation of Salt

Water is one the most powerful and efficient solvent (aka “Universal Solvent”)

- You already know that this power results from water’s **Polar Nature**.
- Solids and gases with relatively strong electrical charge (polar characteristics) will be dissolved.
- Other polar liquids will mix and dilute in water.

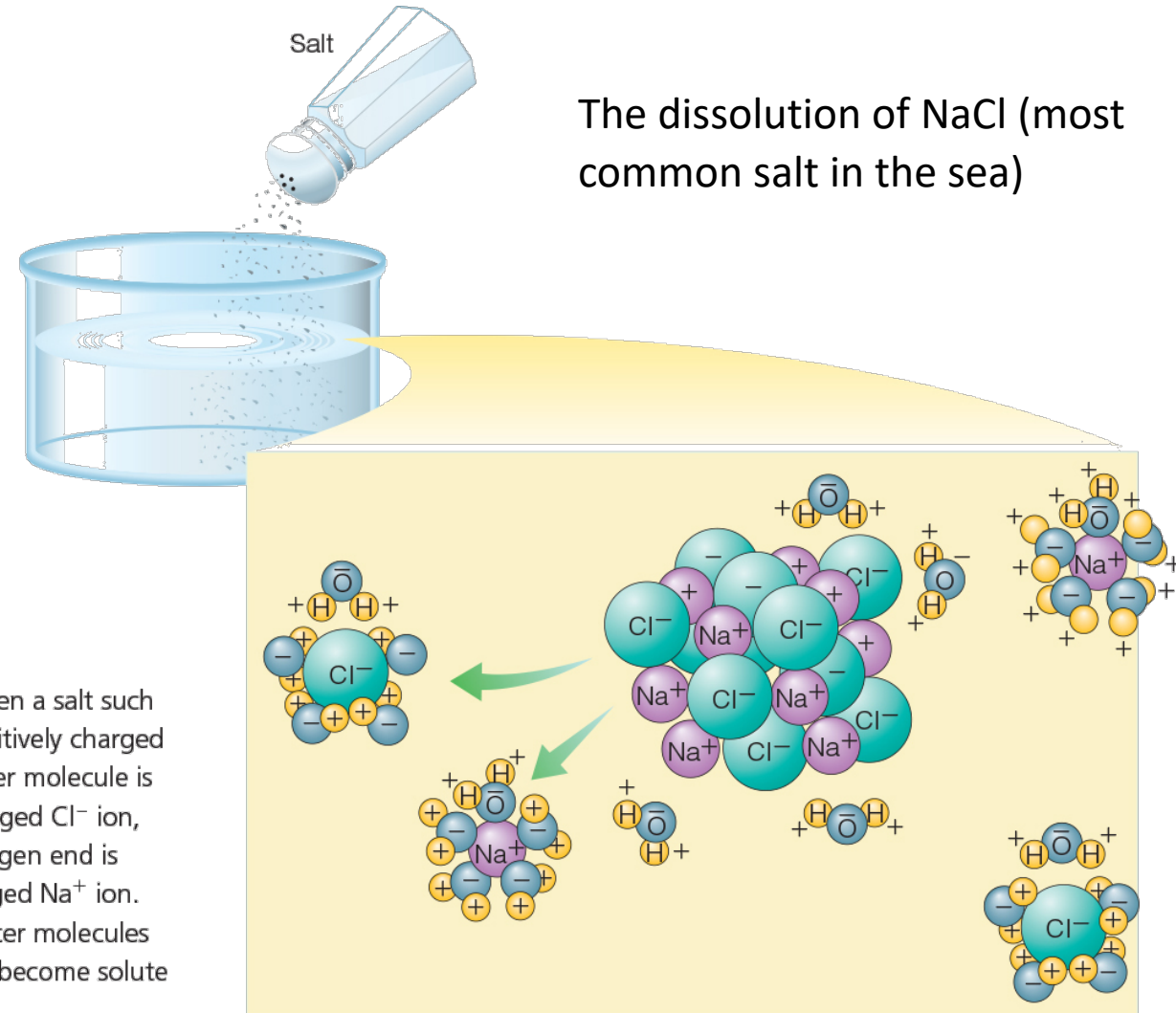


Figure 7.3 Salt in solution. When a salt such as NaCl is put in water, the positively charged hydrogen end of the polar water molecule is attracted to the negatively charged Cl^- ion, and the negatively charged oxygen end is attracted to the positively charged Na^+ ion. The ions are surrounded by water molecules that are attracted to them and become solute ions in the solvent.

Icebreaker Discussion

1. What type of substances is water not able to dilute?
2. What does that mean for humans and life in the ocean?
3. What is meant by the statement “**The Solution to Pollution is Dissolution**”? Do you agree if that statement?
4. Do you think there are environmental impacts of practicing the statement in 3?

Icebreaker Discussion

1. What type of compounds is water not able to dilute?

Nonpolar molecules are hydrophobic and won't dissolve in water.

Ex: Lipids (Fat). Lipids are huge chains of carbon with almost no charge and very strong covalent bonds.

2. What does that mean for humans and life in the ocean?

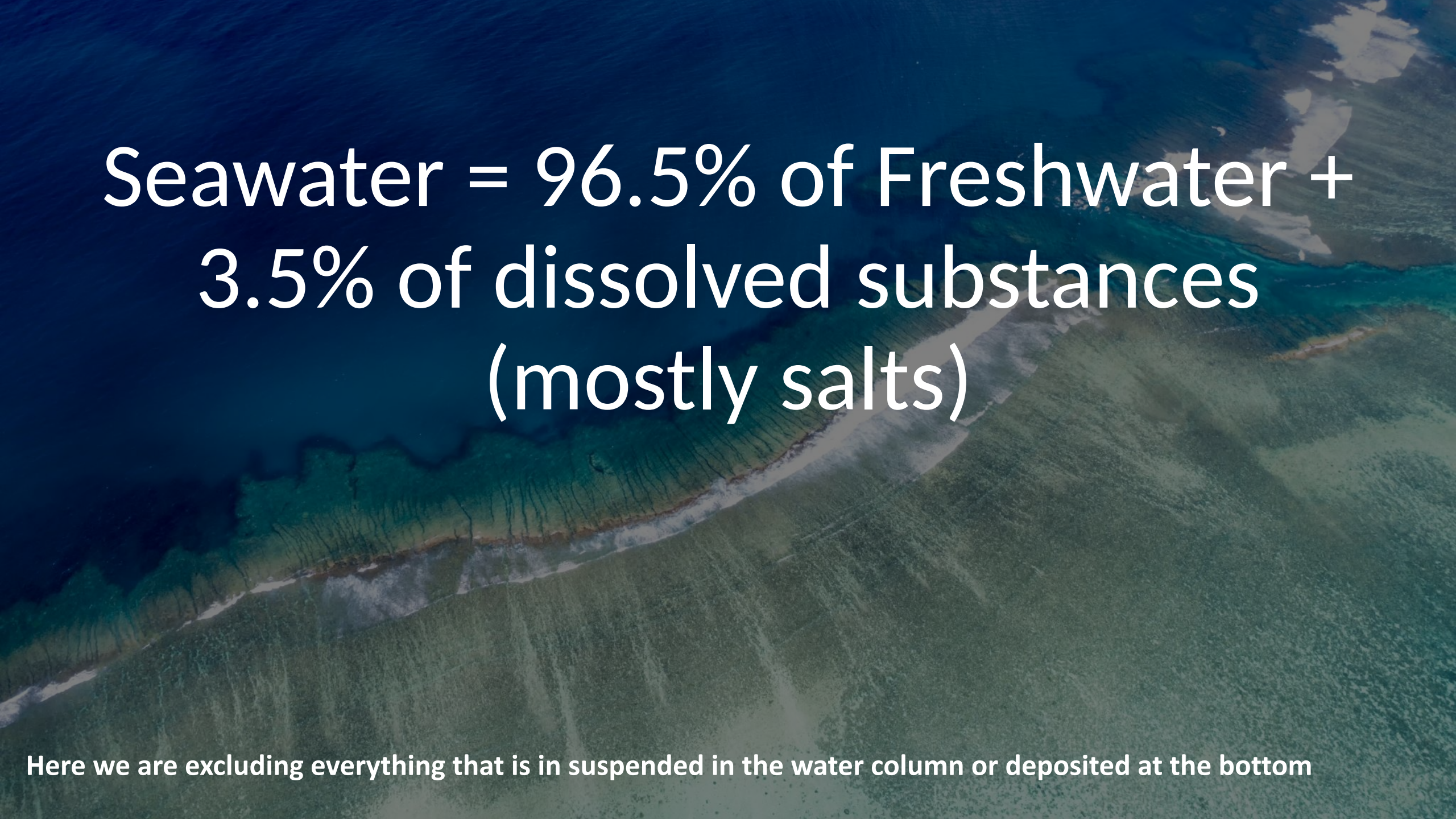
These molecules are important constituents of organic matter. Organisms' cells won't be dissolved in water allowing life thrive in the ocean.

3. What is meant by the statement **“The Solution to Pollution is Dissolution”**? Do you agree with that statement?

If you dilute a pollutant enough, the resultant intensity of the pollution is reduced. By adding clean water to a contaminated product the toxicity of the resultant mixture will be reduced.

4. Do you think there are environmental impacts of practicing the statement in 3?

Some of the environmental impacts that may result from this philosophy are fish kills, algae blooms, elevated bacteria counts, heavy metals accumulating in sediments and marine life. The dilution strategy is “mostly” no longer an acceptable method of dealing with pollution. Scientists and policymakers now understand the need to reduce and eliminate the discharge of pollution into our environment and waterways, not divert it to another location or simply reduce the toxicity of the resultant mixture.

An aerial photograph of a tropical coastline. A narrow white sandy beach runs diagonally across the frame, separating the dark blue ocean from a shallow, turquoise lagoon. The lagoon floor is visible, showing sandy patches and greenish-brown vegetation. The text is overlaid on the dark blue ocean area.

Seawater = 96.5% of Freshwater +
3.5% of dissolved substances
(mostly salts)

Here we are excluding everything that is in suspended in the water column or deposited at the bottom

Major constituents of the seawater

~99.99% of the seawater is freshwater and a few ions

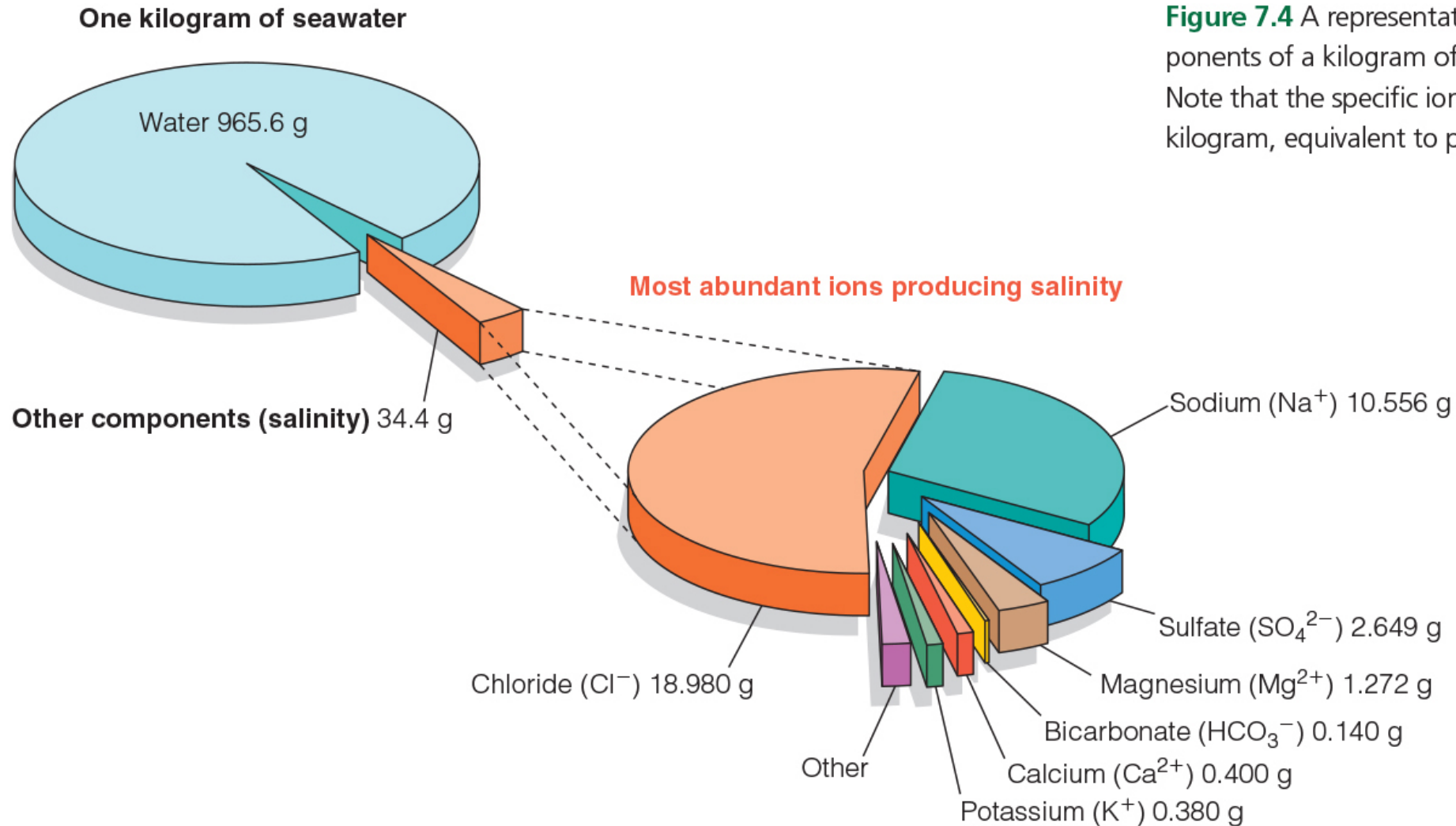


Figure 7.4 A representation of the most abundant components of a kilogram of seawater at 34.4‰ salinity. Note that the specific ions are represented in grams per kilogram, equivalent to parts per thousand (‰).

Major constituents of the seawater

Table 7.1 Major Constituents of Seawater

Constituent	Concentration in Parts per Thousand (‰), or Grams per Kilogram (g/kg)	Percent by Mass	Percent of Total
Water Itself			96.5
Oxygen	857.8	85.8	
Hydrogen	107.2	10.7	
Most Abundant Ions			3.49
Chloride (Cl ⁻)	18.980	1.9	
Sodium (Na ⁺)	10.556	1.1	
Sulfate (SO ₄ ²⁻)	2.649	0.3	
Magnesium (Mg ²⁺)	1.272	0.1	
Calcium (Ca ²⁺)	0.400	0.04	
Potassium (K ⁺)	0.380	0.04	
Bicarbonate (HCO ₃ ⁻)	0.140	0.01	
Total	999.377 g/kg	99.99%	

Source: Data from M.J. Kennish, 1994, cited in *CRC Handbook of Marine Science*, CRC Press, Edited by Walton-Smith, 1994. Table created by Tom Garrison.

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Magnesium (Mg ²⁺)			
Calcium (Ca ²⁺)			
Potassium (K ⁺)			
Bicarbonate (HCO ₃ ⁻)	0.148	0.01	
Total	999.377 g/kg	99.99%	

Note we are missing ~0.01%. Although, these elements (aka Trace Elements) are present in tiny concentrations, they are essential for life!

Trace elements constitute the important nutrients for life in the ocean.

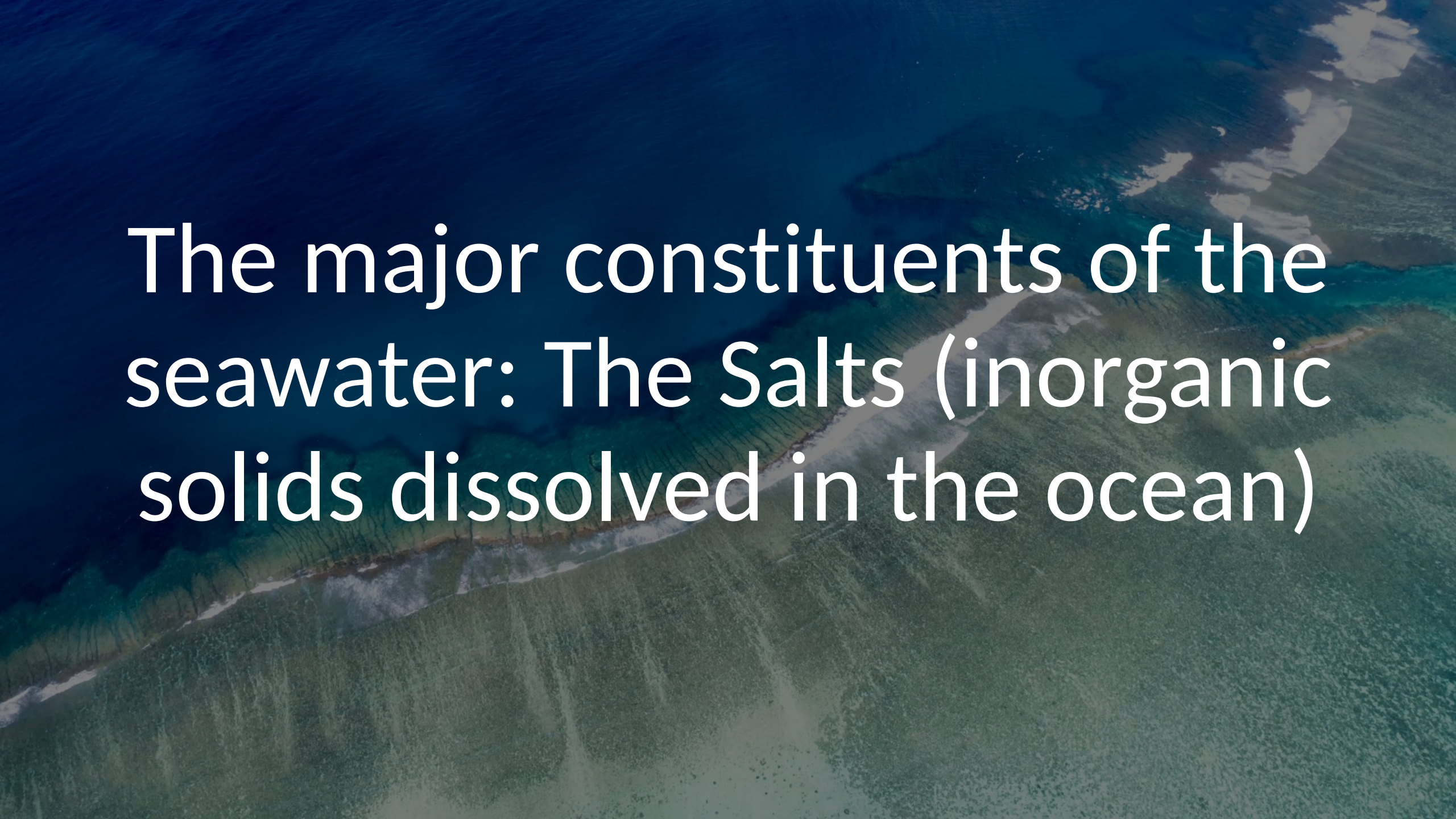
Marine biogeochemistry is the study of the physical, biological, geological, and chemical processes that distribute and transform nutrients and other chemical species (or trace elements) in the marine environment.

Their concentration is so small that it is easier to show them as parts per million (or mg/Kg) or parts per billion (or mg/Tons).

Table 7.2 Minor and Trace Elements in Seawater at 35‰ Salinity

Constituent	Concentration in Parts per Thousand (‰), or g/kg	Concentration in Parts per Million (ppm), or mg/kg (mg/L)	Concentration in Parts per Billion (ppb), or mg/1,000 kg
Minor Elements			
Bromine (Br)	0.065	65	
Strontium (Sr)		8	
Boron (B)		4	
Silicon (Si)		3	
Fluorine (F)		1	
Important Trace Elements			
Nitrogen (N) ^a		0.28	280
Lithium (Li)			125
Iodine (I)			60
Phosphorus (P)			30
Zinc (Zn)			110
Iron (Fe)			6
Aluminum (Al)			2
Manganese (Mn)			2
Lead (Pb)			0.04
Mercury (Hg)			0.03
Gold (Au)			0.0013

Source: Data from M. J. Kennish, 1994, cited in *CRC Handbook of Marine Science*, CRC Press, Edited by Walton-Smith, 1994. Table created by Tom Garrison.
^aRefers to nitrogen available as a nutrient, not as dissolved gas.

An aerial photograph of a tropical coastline. The image shows a series of white sand beaches and turquoise shallow waters (lagoons) separated by dark green, forested land. The water transitions from a light turquoise near the shore to a deep blue further out. The text is overlaid on the upper half of the image.

The major constituents of the seawater: The Salts (inorganic solids dissolved in the ocean)

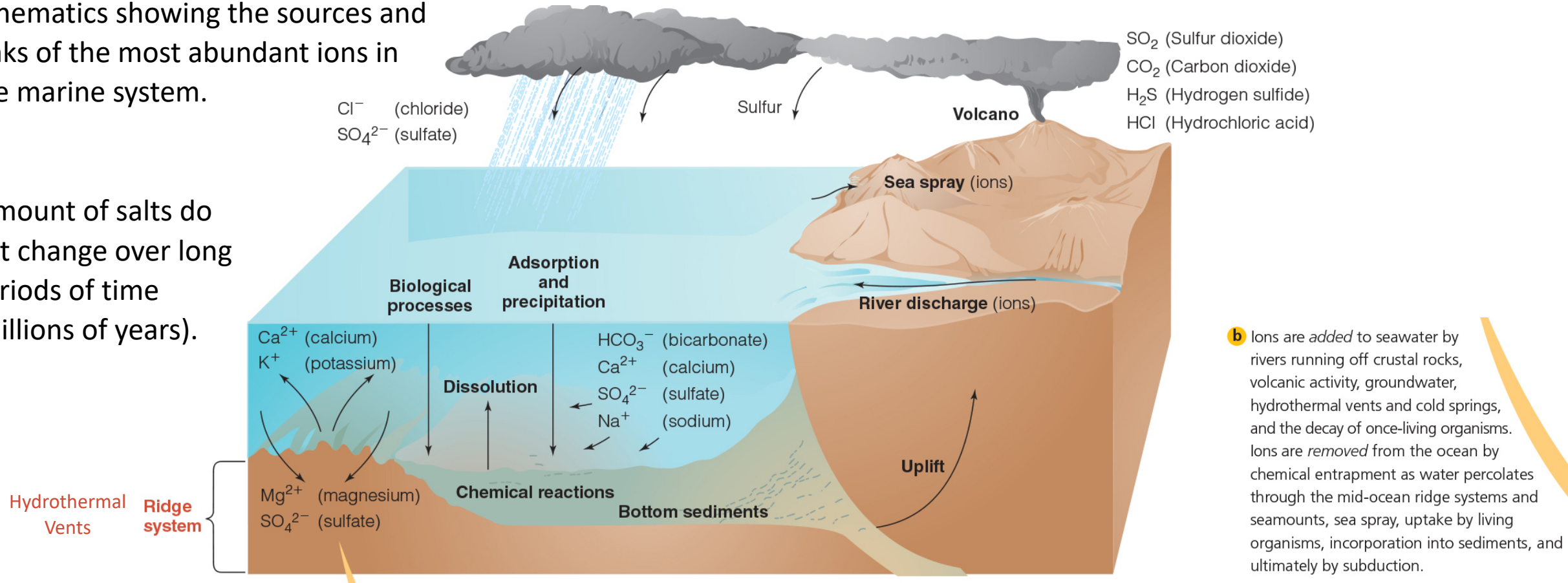
When we studied the formation of ocean basins and salinity, we found out that water and salts must have been accumulating in the depressions of the Earth's crust creating a salty ocean.

1. Where do the ocean salts come from?
2. Is the ocean getting saltier over time?

Processes adding (sources) and removing (sinks) salts in the ocean. Ocean is in an apparent equilibrium.

Schematics showing the sources and sinks of the most abundant ions in the marine system.

Amount of salts do not change over long periods of time (millions of years).



b Ions are *added* to seawater by rivers running off crustal rocks, volcanic activity, groundwater, hydrothermal vents and cold springs, and the decay of once-living organisms. Ions are *removed* from the ocean by chemical entrapment as water percolates through the mid-ocean ridge systems and seamounts, sea spray, uptake by living organisms, incorporation into sediments, and ultimately by subduction.

Interestingly, the ions that consist the most abundant marine salt (NaCl) do not have the same source.

Processes adding (sources) and removing (sinks) salts in the ocean. Ocean is in an apparent equilibrium.

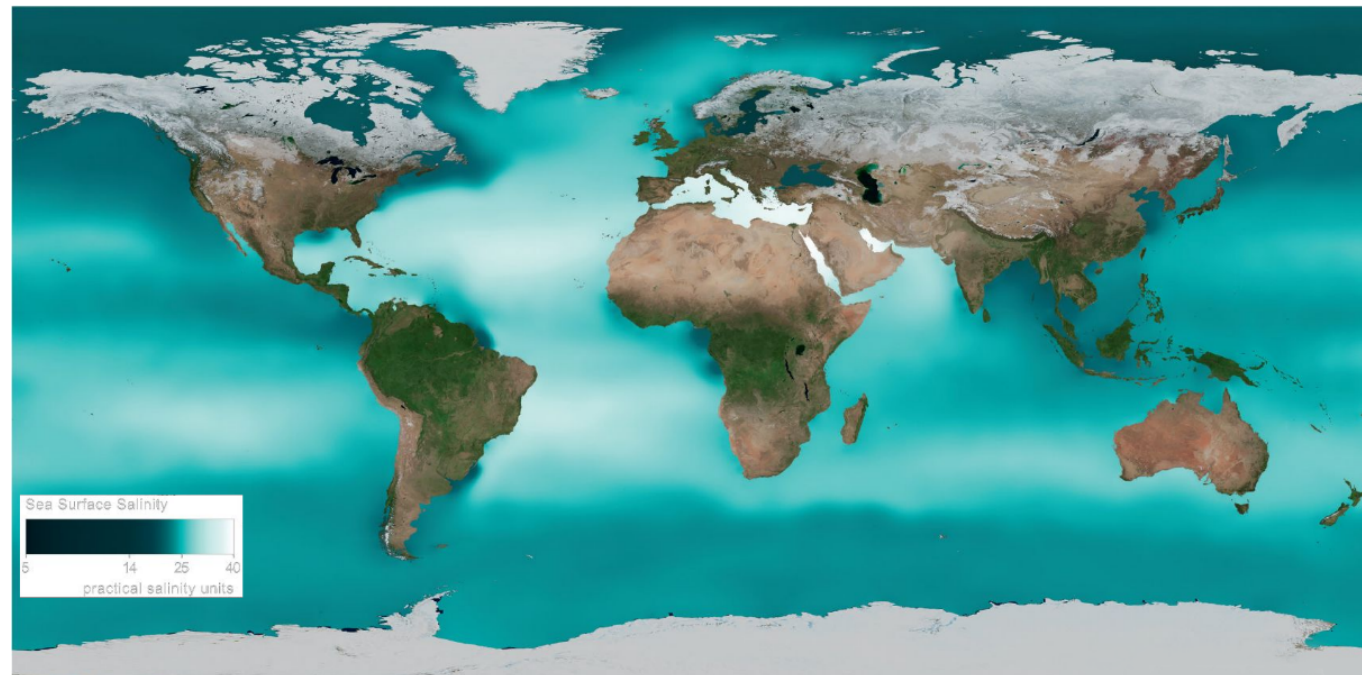
The Ratio of Dissolved Solids in the Ocean is Mainly Constant

- **Forchhammer's principle** (principle of constant proportions)
 - The total amount of dissolved solids(salinity) may vary among samples
 - The ratio of major salts was constant in samples of seawater from many locations
 - Ex: Cl^- will “always” consist 55% of the total solid salts in any seawater samples.
 - Over time we learned this ratio slightly change in different regions.



Georg Forchhammer

Average Ocean Surface Salinity



When we studied the formation of ocean basins, we found out that water and salts must have been accumulating in the depressions of the Earth's crust creating a salty ocean.

1. Where do the ocean salts come from? **Several sources (runoff, dissolution of sediments, volcanic activity, rain)**

2. Is the ocean getting saltier over time?
Not over a long time period, because
Sources ~ *Sinks*

Seawater's conservative and non-conservative constituents

Residence time = Time an element stays in the ocean

$$\text{Residence time} = \frac{\text{Amount of element in the ocean}}{\text{Rate at which the element is added to (or removed from) the ocean}}$$

- Conservative constituents (major constituents) of seawater remain in constant proportions or change very slowly through time - *long residence times*
- Nonconservative constituents (usually trace elements) are those substances dissolved in seawater that are tied to biological or seasonal cycles or to very short geological cycles - *short residence times*

Table 7.3 Approximate Residence Times for Constituents of Seawater

Constituent	Residence Time (years)
Chloride (Cl ⁻)	100,000,000
Sodium (Na ⁺)	68,000,000
Magnesium (Mg ²⁺)	13,000,000
Potassium (K ⁺)	12,000,000
Sulfate (SO ₄ ²⁻)	11,000,000
Calcium (Ca ²⁺)	1,000,000
Carbonate (CO ₃ ²⁻)	110,000
Silicon (Si)	20,000
Water (H ₂ O)	4,100
Manganese (Mn)	1,300
Aluminum (Al)	600
Iron (Fe)	200

**Major (conservative)
constituents of seawater**

Water is obviously not a trace element, but the hydrological cycle is faster than the conservative elements' cycles

**Trace (non-conservative)
constituents of seawater.**

Sources: Data from Broecker and Peng, 1982; Bruland, 1983; Riley and Skirrow, 1975.

Measuring the seawater salinity

Key Concept: Salinity = grams of salt (inorganic solids) dissolved in 1 kg of water (aka parts per thousand ‰)

Measuring the seawater salinity

Oceanographers found out that the conductivity of the water is proportional to the salinity. So, we deploy conductivity sensors and collect water (“conductivity”) samples.

Deployed sensors usually have to be calibrated using a more precise instruments. So, we collect water samples to be analyzed in the lab using a Salinometer (errors are very small $\sim 0.001\%$)

Modern Salinometer



Water sampling system. Conductivity, temperature, and pressure sensors are located below the bottles.

Courtesy of Officers and Crew of NOAA Ship PISCES; Collection of Commander Jeremy Adams, NOAA Corps



Figure 7.6 A “rosette” of sampling bottles. Each of these bottles may be separately closed by a signal from the research ship when the array reaches predetermined depths. The bottles are hauled to the surface and their contents analyzed by a salinometer.

For the curious: The hydrological cycle

Figure 7.2 A simplified hydrologic cycle. Water moves from ocean to air, onto land, to lakes and streams and groundwater, back to the sky and ocean, in a continuous cycle. The numbers indicate the approximate volumes of water in cubic kilometers per year (km^3/yr). Water is also stored in the ocean, ice, groundwater, lakes and streams, and the atmosphere.

